

Scientific Machine Learning: Blending Science with Data

Homework Days 13+14

You are given measurements of the movement of a mass on a spring, forced by an unknown force $r(t)$. You are interested in the motion $t \in [0, 1]$. You are also given that the mass stops moving at $t = 1$, i.e. $dx/dt(t = 1) = 0$. The motion of the mass is dependent on the initial location of the mass $x(t = 0)$. The data for $x(t = 0) = 0$, $x(t = 0) = 1.0$, $x(t = 0) = 2.0$, $x(t = 0) = 0.5$ are stored in files 'DataHomeworkDay13a.csv', 'DataHomeworkDay13b.csv', 'DataHomeworkDay13c.csv', 'DataHomeworkDay13d.csv' respectively. The first column of this file is the value of $x(t = 0)$, second column is t_i and third column is $x_i \approx x(t_i)$. You can assume that t_i is equally spaced and $t_i = 0, 0.02, 0.04, \dots, 1.0$.

- a) Download and plot x vs time for different values of $x(t = 0)$.
- b) (**LEVEL 3**) You would like to find a good Neural Network to predict the data for $x(t = 0) = 0.5$. You employ the following strategies
- Train with the data for $x(t = 0) = 0$ and use it to predict the data for $x(t = 0) = 0.5$
 - Train with the data for $x(t = 0) = 0$ and $x(t = 0) = 1.0$ and use it to predict the data for $x(t = 0) = 0.5$
 - Train with the data for $x(t = 0) = 0$, $x(t = 0) = 1.0$ and $x(t = 0) = 2.0$ and use it to predict the data for $x(t = 0) = 0.5$
 - Train with the data for $x(t = 0) = 0$, $x(t = 0) = 1.0$, $x(t = 0) = 2.0$ and $x(t = 0) = 0.5$ and use it to predict the data for $x(t = 0) = 0.5$
- c) Which of the training strategies give the best prediction of the data with $x(t = 0) = 0.5$? Comment on your observations from your results in a) and b).